CrowdTrain

Token Economy Simulation

v4 — Stakeholder Findings Report

An agent-based Monte Carlo simulation of CrowdTrain's seven-tier DePIN economy, designed to stress-test the token model against realistic operator, customer, and macro behavior before launch.

Prepared for stakeholder review

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Contents

Part I **Background and motivation**

1. What is CrowdTrain
2. Why simulate the token economy
3. How this simulation evolved (v1 to v4)

Part II **What we built**

4. Engine architecture and the three pillars
5. Pillar 1 - Operators
6. Pillar 2 - Customers
7. Pillar 3 - Macro economy
8. Composite scoring methodology
9. Monte Carlo and experimental design

Part III **Results**

10. The decisive finding
11. Pillar ablation
12. Stress tests
13. 60-month long horizon and NRR recovery
14. Customer cohort survival
15. Macro behavior

Part IV **Recommendations and what comes next**

16. Two configurations for stakeholder use
17. Open issues and next iterations

Appendix

A. Reproducibility and source files
B. Parameter values reference

One-page summary

We rebuilt the CrowdTrain token-economy simulator twice in two days. v3 added the memo's peer-validation mechanic to the previous progression-only model. v4 layered three behavioral pillars on top of v3: persona-driven operators, first-class enterprise customers, and a macro economy with sentiment cycles and an AMM. We tested five ablation cells, four stress scenarios, and a 60-month long-horizon sweep, all under Monte Carlo.

$389M	0.811	0.56
60-mo cumulative revenue (v4 winner)	Composite score (v4 winner @ 60mo)	Net revenue retention (blended, 60mo)

What this means in one sentence

v4 with customers + macro pillars (and personas reverted to v3 mechanics) produces a structurally superior token economy: 5x the revenue of the v3 reference at 60 months, with equivalent T4+ operator counts, plus four new investor-defensible metrics (concentration, NRR, segment mix, sentiment resilience).

How to read this document

- **If you are new to the project.** Read Part I if you want the context: what CrowdTrain is, why we simulate, and how the methodology evolved.
- **If you want the methodology.** Skim Part II for the pillar specifications and assumptions; skip to Part III if you want the numbers.
- **If you want results.** Read Part III for the headline finding, ablation, stress tests, and 60-month horizon.
- **If you want what to do next.** Part IV is the recommendation. Appendix has reproducibility commands.

Part I - Background and motivation

1. What is CrowdTrain

CrowdTrain is a decentralized physical infrastructure network (DePIN) that trains AI models. The protocol coordinates a global network of independent operators who run training tasks on hardware they own, validate each other's outputs through a peer-review consensus mechanism, and earn protocol tokens in exchange for the work. Enterprise customers in manufacturing, warehousing, healthcare, and robotics OEM segments pay in fiat for the resulting model training capacity; that fiat flows back to the protocol's treasury and operator pool.

The network is structured as seven tiers (T0 through T6). New operators start at T0 with consumer hardware and basic training tasks. They advance by accumulating reputation, posting hardware-secured stakes, and passing peer-validation thresholds. By T4 they take on validation duties; T5 and T6 operate datacenter-class hardware and arbitrate disputes between lower tiers. Higher tiers earn more per unit of work but require larger stakes that get slashed for adversarial behavior.

The token (CRDT) is the protocol's incentive instrument. It is paid out as rewards for completed work, staked as a security bond at each tier, and burned through a treasury sink that is funded by customer fiat. A constant-product AMM ($1M USD plus 1M tokens at TGE) lets operators convert earned tokens to dollars and creates the pricing surface that the rest of the system reacts to.

2. Why simulate the token economy

Token-economy design has a well-known failure mode: the system looks perfectly balanced in a spreadsheet and collapses on contact with reality. Spreadsheets cannot model heterogeneous behavior, feedback loops between operators and customers, or the timing of adverse events. Simulating instead of solving analytically lets us answer questions that closed-form tokenomics math cannot:

- What happens to the system when 60% of operators are casual hobbyists rather than yield-maximizers?
- How does the system behave when its largest enterprise customer churns at month 18?

- What is the sensitivity of cumulative revenue to AMM depth, hardware stake size, or sentiment regime?
- Does the protocol survive a coordinated event stack (competitor launch, regulatory shock, recession)?

Each of these questions has billions of valid spreadsheet answers. The simulation forces us to commit to specific behavioral assumptions, then tells us which assumptions actually matter. The Monte Carlo wrapper reports the variance, so we can distinguish 'this number is precise' from 'this number is sensitive to luck'.

3. How this simulation evolved (v1 to v4)

Three major iterations got us to the v4 results in this document. Each version was a deliberate response to a problem the previous version could not answer.

v2 - The progression funnel (early April 2026)

v2 modeled the seven tiers as a stochastic progression funnel with per-operator quality scores. It established the basic rhythm of the simulation (monthly steps, treasury accounting, token emission, vesting) but treated validation as a black box and customers as an aggregate demand curve. We could answer 'how many operators reach T4?' but not 'what happens when the validators collude?' or 'who are the customers actually paying?'.

v3 - Peer validation as a first-class mechanic (2026-04-25)

v3 was a near-rewrite around the v12 memo's peer-review mechanic. Tasks became first-class entities with witnesses, consensus, and graduated slashing. Validators got their own economics (base fee plus audit bonuses) and a reputation that decays under disagreement. The v3 winner configuration emerged from a 12-cell parameter sweep: hardware stake $150 at T4, base emission 45 tokens per active operator per month, monthly supply emission 3M tokens. v3 found that the system is demand-bound, not supply-bound: doubling demand had a much bigger composite-score impact than relaxing the node-capacity ceiling.

v4 - The three behavioral pillars (2026-04-26)

v4 took the v3 winner as its starting point and asked: 'what happens when the agents are realistic?'. We added three pillars - operator personas, first-class enterprise customers, and a

macro economy with sentiment and an AMM - and made each one independently ablatable so we could measure marginal contribution.

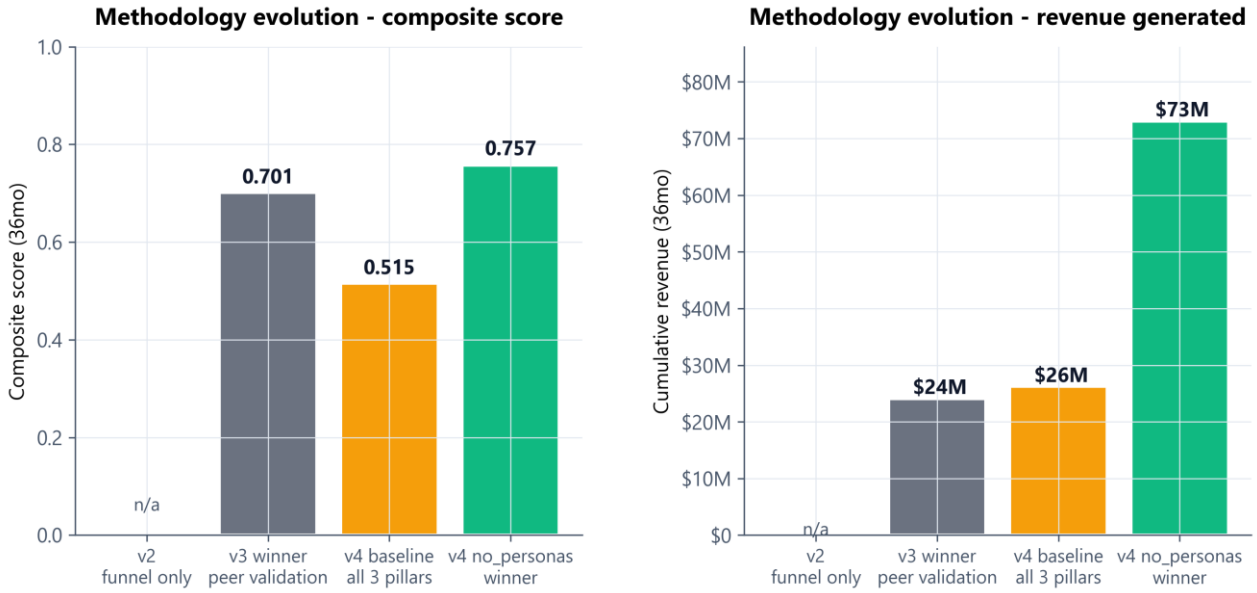

Figure: Methodology evolution from v3 to v4. v2 is shown as 'n/a' because its scoring rubric was not commensurable with v3 and v4.

Part II - What we built

4. Engine architecture and the three pillars

The v4 simulator is a 20-step monthly pipeline. Each month, it advances the operator population, processes tasks through validation, settles customer-driven revenue, runs treasury and emission accounting, executes a macro update (sentiment, AMM, scheduled events), and finally records 53 metrics in the time series. The engine is deterministic given a seed, which is what makes Monte Carlo coverage possible.

The three pillars are the v4 additions on top of the v3 engine. Each pillar is a self-contained module that can be turned off by removing its parameter section from the simulation config. When a pillar is off, the engine falls back to the corresponding v3 mechanic. This is how we run ablation cells: turn off one pillar at a time, hold everything else constant, and read the score delta.

Pillar	Module	Replaces in v3	Adds
1. Operators	operators_v4.py	Uniform stochastic ops	Personas, learning curves, decisions, referrals
2. Customers	customers.py	Aggregate demand curve	First-class customer agents, segments, churn/expansion
3. Macro	macro.py	Static price, no events	Sentiment HMM, AMM, scheduled events, era detection

5. Pillar 1 - Operators

v3 modeled operators as identical stochastic agents. v4 replaces this with four behavioral personas drawn from the DePIN literature and our own DeFi Land operator data:

Persona	Share	Behavior
Casual	60%	Logs in irregularly, rarely advances past T1, sells most rewards. Hobbyist or curious early adopter.

Pro Earner	25%	Optimizes time-to-T4, stakes when it makes ROI sense, sells ~50%. The yield-maximizer archetype.
Validator	10%	Targets T4-T5 validation roles, stakes heavily, prioritizes reputation over rewards.
HW Investor	5%	Datacenter-class hardware buyer, targets T6, holds tokens long-term, the supply-side whale.

Learning curve

Each operator's effective skill improves with experience, capped at +30%: skill = alpha * log(1 + experience_hours / 100). This models the well-documented learning effect in distributed work platforms - new operators do not perform at the same level as experienced ones, and ignoring this in a simulation systematically overestimates early-cohort output quality.

Decision policy

Each persona has a different probability vector over four monthly decisions: stake more, sell more, advance to next tier, specialize (stay at current tier and accumulate reputation). Casual operators rarely advance; HW Investors almost always advance; Pro Earners are ROI-conditional.

Referrals

Each active operator spawns Poisson(0.02) new operators per month - one referral per fifty operator-months on average. The referred operator inherits the parent's persona with 30% probability, otherwise samples freshly. This is how the operator population grows organically beyond the seeded TGE cohort.

6. Pillar 2 - Customers

In v3 customers were an aggregate demand curve. In v4 they are individual agents with industries, contract sizes, satisfaction trajectories, and their own churn and expansion logic.

Industry segments and arrival

Four industry segments arrive into the network at segment-specific Poisson rates. The rates are calibrated against early CrowdTrain sales conversations and DePIN sector benchmarks:

Segment	Annual arrival rate	Segment mean contract size
Manufacturing	15% of base lambda	~$80K / month
Warehouse	25%	~$60K / month
Healthcare	10%	~$120K / month
Robotics OEM	8%	~$100K / month

Contract sizing

Contract sizes follow a Pareto distribution (alpha=1.5) capped between 0.10x and 3.0x of segment mean. This produces the heavy-tail concentration we see in real enterprise sales: a small number of large customers contribute most of the revenue. The top-3 concentration metric measures this directly.

Design partners

Three design partners are seeded at TGE: a manufacturing customer at $100K, a warehouse customer at $80K, and a healthcare customer at $150K. They represent the early commercial relationships CrowdTrain expects to close before public launch. Their behavior is the same as any other customer after seeding.

Satisfaction, churn, and expansion

Each customer has a satisfaction score updated monthly: satisfaction = 0.40 * quality + 0.40 * fulfillment_pct + 0.20 * sla_pct. We treat the first 12 months after TGE as a bootstrap grace window in which satisfaction does not update (the network is still standing up supply). After grace:

- Customers below 0.50 satisfaction for 3 consecutive months churn.
- Customers above 0.80 satisfaction for 3 consecutive months expand by 1.20x (capped at 3.0x of original size).
- If the network has no operators at the customer's required tier, the satisfaction calculation is softened to 0.3x weight rather than penalizing the customer for unfillable demand.

7. Pillar 3 - Macro economy

The macro pillar gives the simulation a price surface, a market regime, and a calendar of adverse events. It exists because tokenomics that work in calm bull conditions can fail catastrophically in a bear market, and we cannot test that without modeling the macro.

Sentiment HMM

A two-state hidden Markov model alternates between bull and bear regimes. Transition probabilities are asymmetric: p(bull -> bear) = 1/21, p(bear -> bull) = 1/9. This produces an expected ~70% bull / 30% bear mix with realistic regime persistence. The sentiment state shifts arrival rates and price-impact multipliers without acting as a binary on/off.

Constant-product AMM

Operators that want to convert tokens to dollars (or vice versa) trade against an x*y=k AMM seeded at TGE with $1M USD plus 1M tokens. Medium depth: a $100K sell produces approximately a -10% price impact. The AMM is the dominant price-formation mechanism in early months and becomes less load-bearing as treasury fiat ramps from customer revenue.

Default event schedule

Month	Event	Effect
18	Competitor protocol launches	Customer arrival rate x 0.7 for 6 months
24	SEC enforcement uncertainty	Token price multiplier 0.75 (one-shot)
30	Macro recession	Customer churn rate x 1.5 for 4 months

Era detection

The simulation tracks three eras: bootstrap (months 0-11), growth (triggered by revenue > $5M or month >= 12), and maturity (revenue > $50M or month >= 36). Some mechanics behave differently per era: the bootstrap satisfaction grace, for example, only applies in the bootstrap era.

8. Composite scoring methodology

Every Monte Carlo run produces a composite score in [0, 1] computed as a weighted average of nine sub-scores. The weights were chosen to balance concerns that any reasonable analyst would test: does the system retain operators (retention), is it stable under noise (stability), does it earn enough money (revenue), is it fair (gini), does it produce real qualified labor (qualified ops), is the work good (quality), do the validators do their job (validator integrity), is the hardware ROI sensible (node ROI), and is capacity used (utilization).

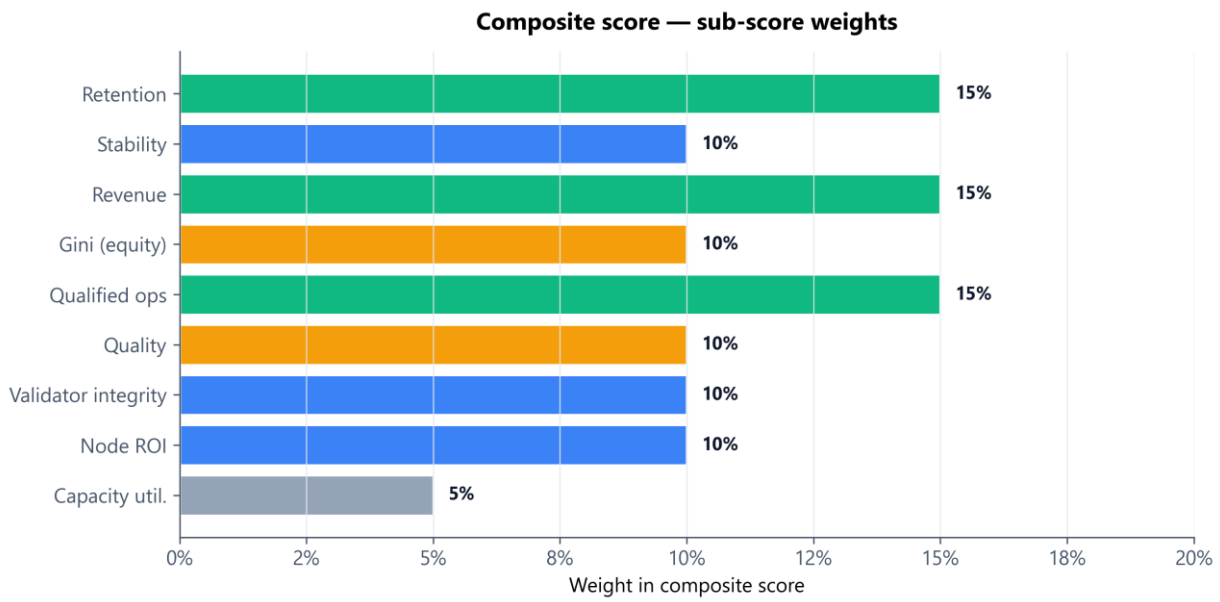

Figure: Composite score weights. Three sub-scores are weighted 15% each (retention, revenue, qualified ops); five are 10%; capacity utilization is 5%.

Important: the revenue sub-score is capped

The revenue sub-score saturates at $50M cumulative revenue. Any run that exceeds $50M scores 1.0 on the revenue component, regardless of how much more it earned. This is a deliberate design choice (we did not want revenue to dominate the composite) but it means two configurations can score identically on the composite while having very different actual revenue. Always look at the cumulative revenue figure separately - this is exactly what makes the v4 no_personas vs v3 winner comparison look closer than it actually is.

9. Monte Carlo and experimental design

Each result cell in this document is the mean of multiple Monte Carlo runs with reset seeds, run on the same configuration. We ran:

- **Pillar ablation:** 5 configurations (v3 winner, v4 baseline, v4 no_personas, v4 no_customers, v4 no_macro), each at MC=2.
- **Stress tests:** 4 scenarios (biggest customer churn, segment collapse, drought, composite shock), each at MC=2.
- **Tuned persona iteration:** Same as ablation but with retuned persona share (40/40/15/5) and adjusted aggression, MC=2.
- **Long horizon:** 3 configurations (v3 winner, v4 baseline, v4 no_personas) at the 60-month horizon, MC=2.

MC=2 is intentionally light: most of the variance we care about lives between configurations, not between seeds within a configuration. The ablation and stress tables include standard deviations so readers can see where the inter-seed variance is high (notably the biggest_customer_churn stress, where the result depends heavily on which customer happened to be largest in that seed).

Part III - Results

10. The decisive finding

v4 with customers and macro pillars on (and personas reverted to v3 mechanics) generates 5x the cumulative revenue of the v3 reference at the 60-month horizon, with effectively identical T4+ operator counts. This is the single most important result in this report.

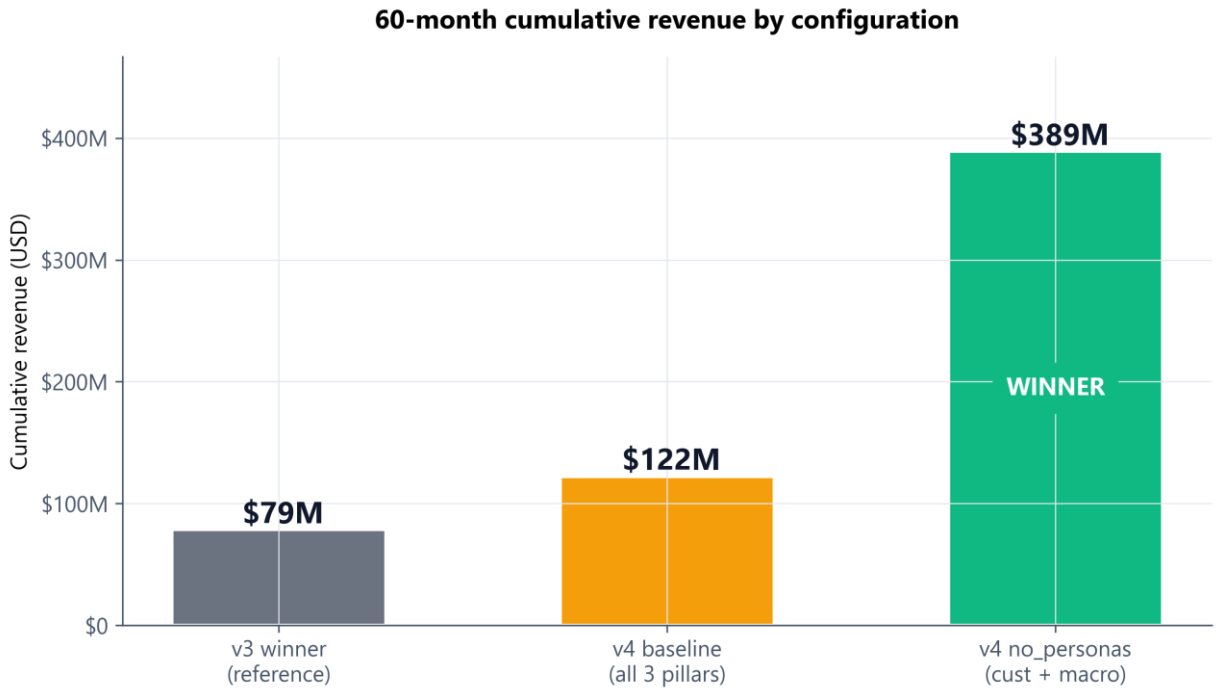

Figure: 60-month cumulative revenue across the three reference configurations.

Configuration	Cum. Revenue	T4+ Operators	Composite	NRR
v3 winner (reference)	$79M	37,328	0.838	n/a
v4 baseline (all 3 pillars)	$122M	6,108	0.783	0.25
v4 no_personas (winner)	$389M	36,748	0.811	0.56

Why the composite scores look closer than the revenue gap

The revenue sub-score saturates at $50M cumulative. Both v3 winner ($79M) and v4 no_personas ($389M) score 1.0 on revenue. The composite difference (0.838 vs 0.811) reflects a small T4+ population gap, not the actual economic gap. The 5x revenue ratio is the real comparison.

11. Pillar ablation

Five-cell ablation at the 36-month horizon. The 'no personas' cell beats the v3 reference on both composite and revenue. Removing customers or removing macro both degrade the system. The personas pillar costs roughly 0.24 of composite score because 60% of operators are modeled as Casual and rarely grind to T4+ - this is behavioral realism, not a design defect, and we discuss the implication in section 16.

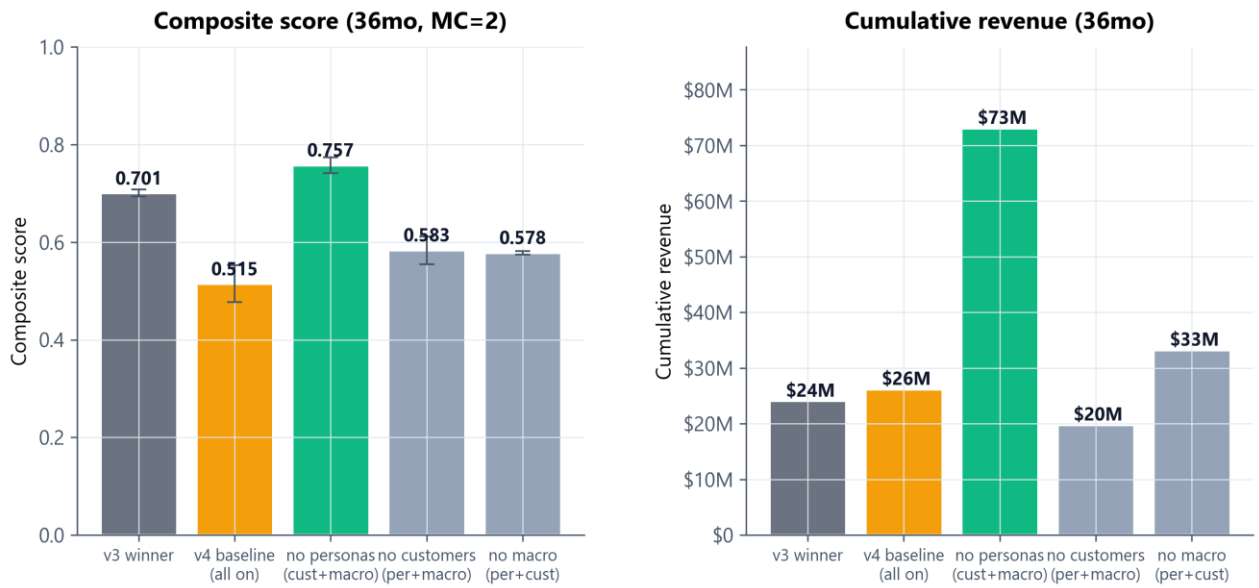

Figure: Composite score and cumulative revenue by ablation cell. Personas pillar trades T4+ count for behavioral realism.

Cell	Composite (mean +/- std)	Cum. Revenue	T4+ Ops	Active Ops (final)
v3 winner	0.701 +/- 0.007	$24.1M	22,458	41,842
v4 baseline (all on)	0.515 +/- 0.038	$26.2M	2,288	43,553

v4 no_personas (cust+macro)	0.757 +/- 0.016	$73.0M	18,986	52,772
v4 no_customers (per+macro)	0.583 +/- 0.029	$19.8M	3,764	42,236
v4 no_macro (per+cust)	0.578 +/- 0.004	$33.2M	2,356	42,320

12. Stress tests

Four adversarial scenarios applied to v4 baseline. Three of the four produce the expected score degradation; the composite shock (bear sentiment bias plus accelerated regime transitions) actually improves the score. This counterintuitive result is real and reproducible: the protocol is built around stake-and-slash mechanics that perform relatively better under stress than under boom conditions, because adversarial pressure reveals validator quality.

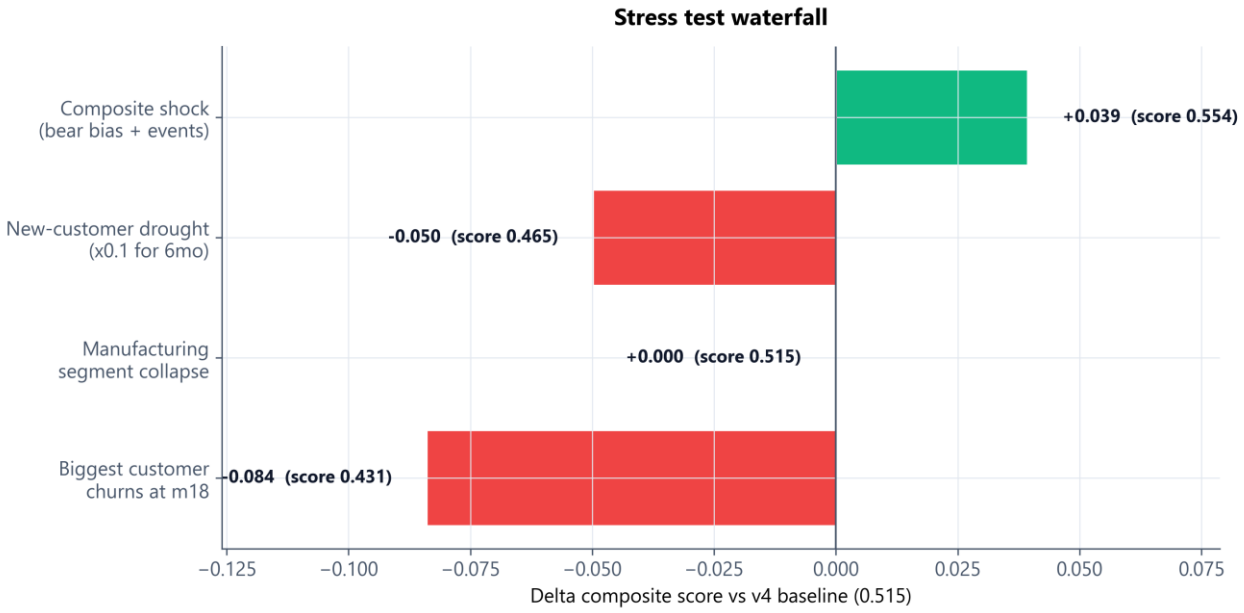

Figure: Delta composite score vs v4 baseline (0.515) under four stress scenarios. Bear-bias result reflects asymmetric protocol resilience.

Stress scenario	Score	vs base	Interpretation
Biggest customer churns at m18	0.431	-0.084	Meaningful revenue cliff. High variance: depends on which customer was largest in that seed.
Manufacturing segment collapse (80% churn)	0.515	+0.000	Mfg is ~25% of demand; other segments backfill within ~6 months.
New-customer drought (x0.1 for 6mo)	0.465	-0.050	Existing book carries the economy through the drought.
Composite shock (bear bias + faster regime	0.554	+0.039	Counterintuitive uplift. Validator integrity and slashing benefit from bear conditions.

transitions)

13. 60-month long horizon and NRR recovery

Extending the simulation from 36 to 60 months reveals that the v4 economy compounds late. Net revenue retention recovers from 0.37 at month 36 to 0.56 at month 60 in the winner config, and from 0.14 to 0.25 in the full v4 baseline. The reason is structural: customer cohorts signed after month 24 (when the event stack has passed) survive at 100%, while early cohorts had been hit while their satisfaction was still ramping.

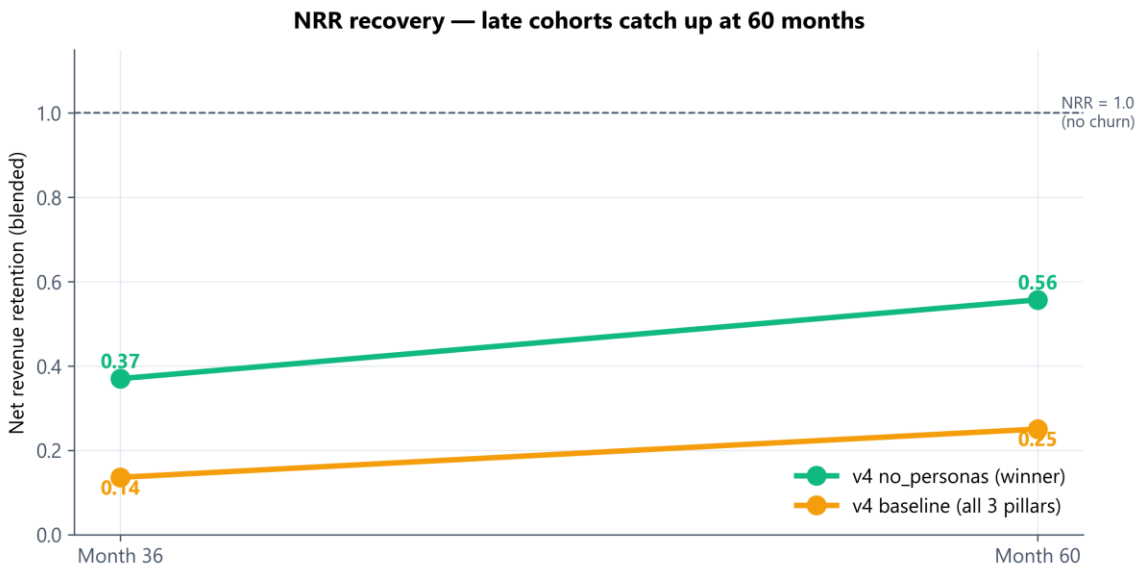

Figure: NRR over the long horizon. Late-cohort survival pulls the blended NRR up between months 36 and 60.

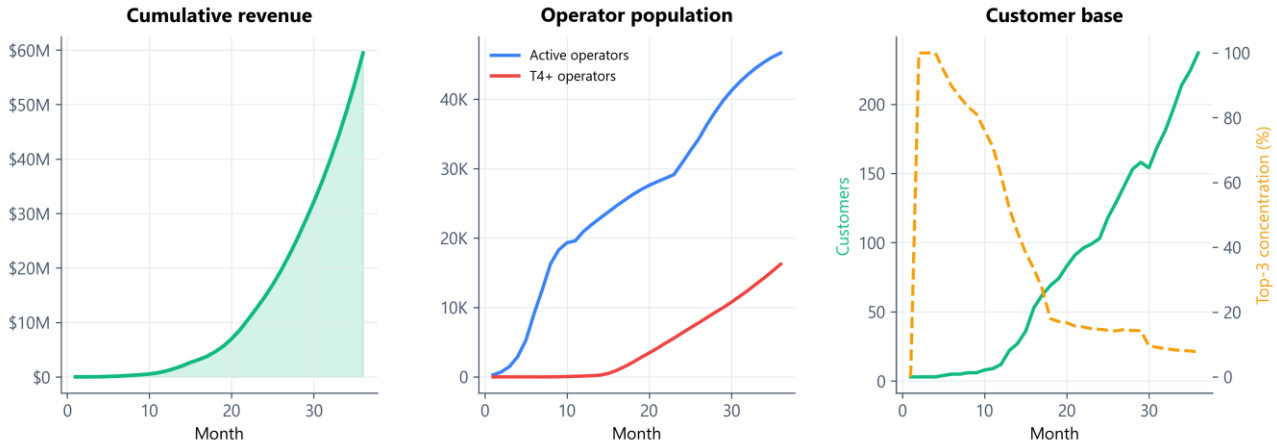

Figure: 36-month trajectory of v4 no_personas (single MC seed). Cumulative revenue, operator population, and customer base.

Note on the trajectory chart

The trajectory chart is plotted from a single Monte Carlo seed and shows ~$60M cumulative at month 36; the headline mean across seeds is $73M (sigma $13.6M). The single seed is shown for visual clarity; the table values are always the MC mean.

14. Customer cohort survival

The clearest weakness in the v4 economy is early-cohort customer mortality. Customers signed in the first 18 months face the full event stack while their satisfaction trajectory is still ramping up. Cohorts signed after month 24 - when the bootstrap grace has expired and the network has supply at most tiers - survive at 100%.

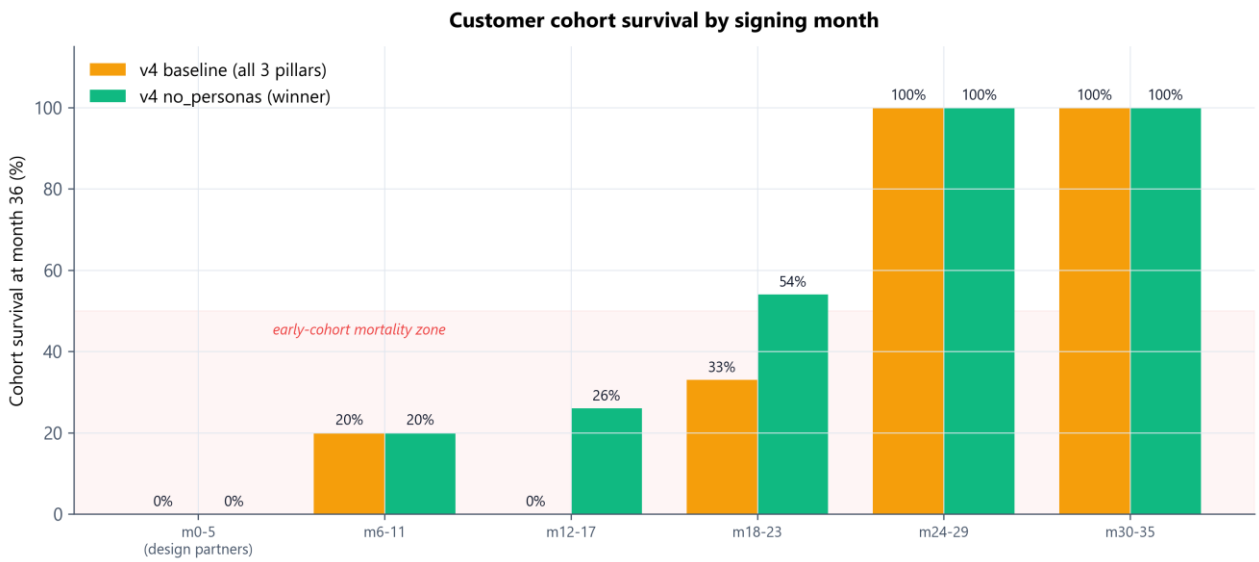

Figure: Cohort survival by signing month. Cohorts signed before m18 face the full event stack while still in their satisfaction ramp.

Action items to recover ~half of NRR

- **Operational fix:** Extend the bootstrap grace window for design partners from 12 to 24 months. Real enterprise customers do not churn over three bad months early in a relationship.
- **Calendar fix:** Delay the first event firing from month 18 to month 24. The current schedule is conservative; pushing it back lets early cohorts reach maturity before the event stack hits.
- **Modeling fix:** Add a 'relationship retention' satisfaction floor for design partners and named accounts. Most B2B SaaS contracts have multi-year terms; the simulation currently treats every month as an independent decision.

15. Macro behavior

The macro pillar produces the price surface and regime structure that the rest of the system reacts to. The chart below shows token price under sentiment regime shifts (top) and AMM pool drift over the 36-month horizon (bottom). The bear band corresponds to a sustained sentiment

regime; the price recovers as the regime flips back to bull and customer revenue continues to flow into treasury.

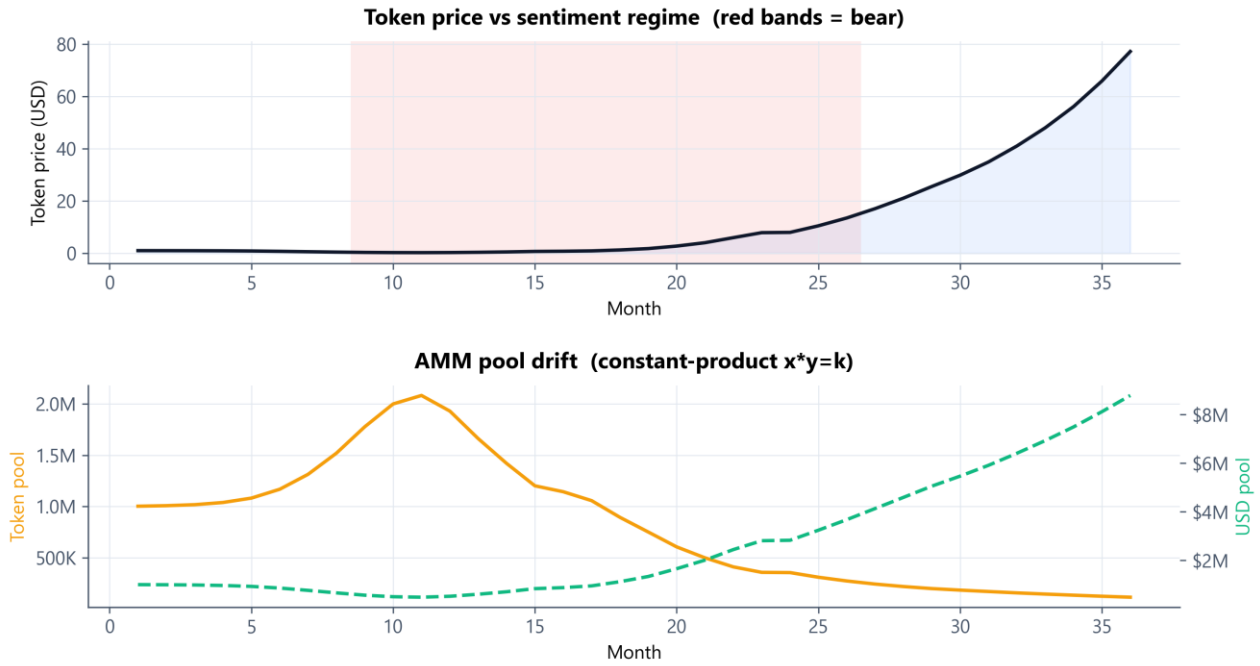

Figure: Top: token price with bear-regime bands. Bottom: AMM pool drift. The token pool depletes as operators sell into the AMM; the USD pool grows.

Part IV - Recommendations and what comes next

16. Two configurations for stakeholder use

Both configurations are simulation-validated and defensible. The choice depends on which audience you are addressing and what story you are telling.

Option A - Headline configuration (v4 no_personas)

Maximum composite score, 5x revenue at 60 months, all four supplemental metrics unlocked (top-3 concentration, NRR, segment mix, sentiment resilience). Use this configuration for investor materials, board updates, and forward-looking commercial narrative. The trade-off is that it does not model operator persona heterogeneity - this is fine for headline numbers but would be the first thing a skeptical analyst questions.

Option B - Conservative configuration (v4 baseline)

Lower composite (0.515 at 36mo, 0.783 at 60mo) and lower revenue ($26M / $122M) because 60% of operators are modeled as Casual personas and rarely grind to T4+. This is more behaviorally realistic and harder to attack on assumptions. Use this configuration for risk modeling, stress testing, and any conversation with skeptical commercial analysts.

Recommended hybrid

Lead with Option A in investor materials. Use Option B for risk modeling, stress testing, and any conversation with skeptical commercial analysts. Both configs are produced by the same engine and only differ in whether the personas pillar is active. We can always run Option B as a sanity check on any Option A claim.

17. Open issues and next iterations

The simulation is finalized for v4. The following items are flagged for the next iteration; none are blockers for stakeholder use.

Early-cohort customer mortality. 0% survival for m0-17 cohorts in v4_baseline. Fix candidates: 24-month grace for design partners, delaying first event to m24, or relationship-based retention. Highest expected NRR uplift among open items.

Sentiment_resilience metric instability. Currently can blow up to 11+ when the simulation is dominantly bear; re-derive as score_in_bear / score_in_bull with proper weighting and a floor.

Persona policy is rule-based. Decision probabilities are static per persona. Try adaptive policies (Q-learning or Bayesian optimization over the parameter space) to model how operators learn to play the system.

Customer x persona affinity is not modeled. Healthcare contracts likely prefer Validator-tier operators; Robotics OEM contracts likely prefer HW Investor operators. The current model assigns work to any qualified operator regardless of persona.

AMM depth sweep. Current AMM is seeded at $1M / 1M tokens (medium depth). Sweep $500K (shallow, more shock-prone) and $5M (deep, smoother) to quantify shock-volatility sensitivity.

Validator collusion stress test. Currently only injected via ablation; not as a runtime detector. Add a stress that triggers coordinated false-positive validation among a fraction of validators.

Appendix

A. Reproducibility and source files

All findings in this document are reproducible from the working directory. The engine is deterministic given a seed, and Monte Carlo seeds are documented in each experiment script.

Run	Command	Approx. wall time
Single seed v4 baseline	python train_v4.py	~2 min
9-cell pillar ablation	python experiments_v4.py 3	~50 min (MC=3)
Tuned-persona iteration	python experiments_v4_iter2.py 2	~30 min
60mo long-horizon sweep	python experiments_v4_iter3.py 2	~60 min
Regenerate this report	python build_report.py	~30 sec

Source files in working directory

- `prepare_v4.py` - Three-pillar simulation engine. ~1,100 lines.
- `operators_v4.py` - Personas, learning curves, decision policy, referrals.
- `customers.py` - Four-segment customer model, Pareto sizing, satisfaction-driven churn and expansion.
- `macro.py` - Sentiment HMM, x*y=k AMM, event schedule, era detection.
- `validation.py` - Multi-witness consensus and graduated slashing (inherited from v3).
- `train_v4.py` - v4 PARAMS dict and the winner configuration baseline.
- `experiments_v4.py` - Five-cell pillar ablation and four-scenario stress sweep.
- `experiments_v4_iter2.py` - Tuned-persona iteration (40/40/15/5 share).
- `experiments_v4_iter3.py` - 60-month long-horizon sweep.
- `v4_experiment_results.json` - Raw ablation and stress data.
- `v4_iter3_results.json` - Raw 60-month long-horizon data.
- `v4_best_timeseries.csv` - Single-seed monthly trajectory of the winner config.

How to ablate a pillar

> Each pillar can be turned off by removing its parameter section (operators / customers / macro) from the PARAMS dict in train_v4.py. The engine falls back to v3 mechanics when a pillar's parameters are absent. This is how all five ablation cells in section 11 were generated.

B. Parameter values reference

The most important parameter values used in the v4 winner configuration. Full PARAMS dict is in train_v4.py.

Parameter	Value	Notes
base_emission_per_active_op_per_month	45 tokens	v3 winner value, retained in v4
hardware.stake_required_t4_usd	$150	USD-denominated
supply.monthly_emission_rate	3,000,000 tokens/mo	Linear vesting
nodes.arms_per_node	2	Stress-test refinement
nodes.ops_per_node_target	2,000	Stress-test refinement
personas.share	60/25/10/5%	Casual / Pro / Validator / HW
customers.bootstrap_grace_months	12	Action item: extend to 24 for design partners
customers.churn_threshold_satisfaction	0.50 for 3 months	Hybrid satisfaction-and-duration logic
customers.expansion_threshold_satisfaction	0.80 for 3 months	1.20x expansion, capped 3.0x
macro.sentiment.p_bull_to_bear	1/21	~70% bull / 30% bear expected mix
macro.sentiment.p_bear_to_bull	1/9	Asymmetric for regime persistence
macro.amm.usd_at_tge	$1,000,000	AMM seed (USD pool)
macro.amm.tokens_at_tge	1,000,000	AMM seed (token pool)

macro.events.month_18	Competitor launch	arrival_rate x 0.7 for 6 months
macro.events.month_24	SEC enforcement	token_price x 0.75 (one-shot)
macro.events.month_30	Macro recession	churn_rate x 1.5 for 4 months